

Definition

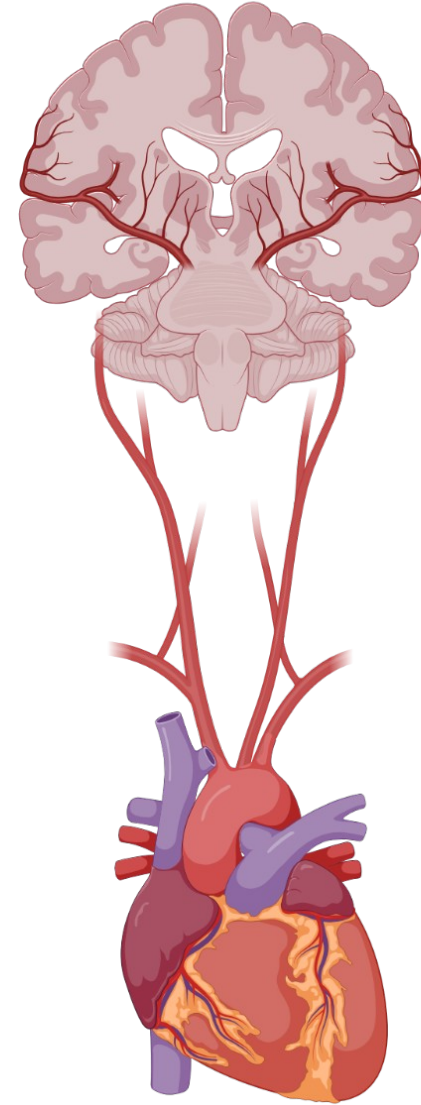
Classification

Evaluation

Management

Cases

Syncope



What is Syncope?

What is syncope?

Transient loss of consciousness (TLOC) due to global cerebral hypoperfusion followed by spontaneous full recovery.

What are some mimics?

LOC + shaking
episodes + post-
ictal state

LOC + vertigo,
dysarthria

LOC preceded by
diaphoresis,
tremulousness

Definition

CPP

Baro-R Reflex

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Physiology of Cerebral Perfusion

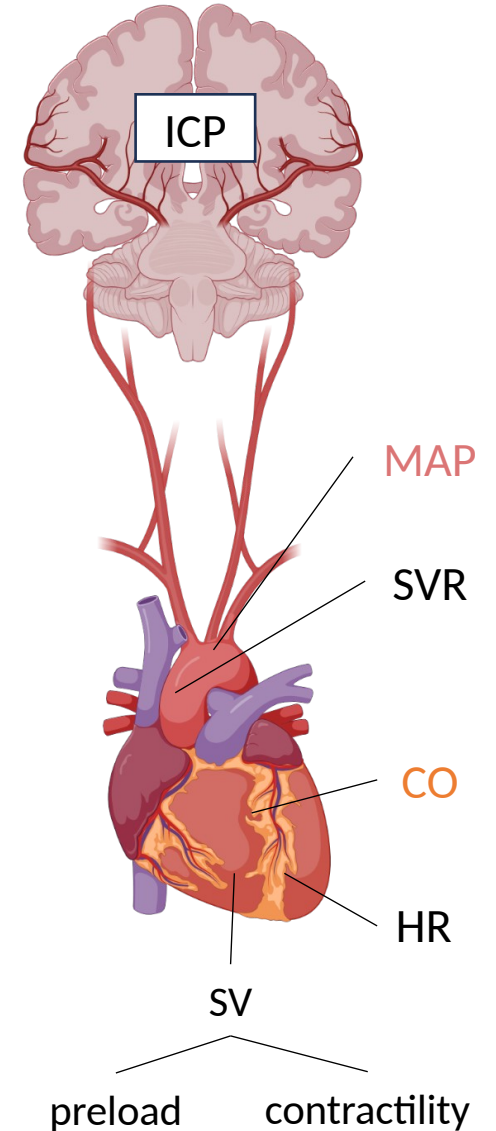
↓ global cerebral perfusion → syncope

CPP (cerebral perfusion pressure)
= MAP - ICP

MAP = CO x SVR

CO = HR x SV

SV is impacted by preload,
afterload (SVR) and contractility



Definition

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What determines
cerebral perfusion?

What determines mean
arterial pressure
(MAP)?

What determines
cardiac output (CO)?

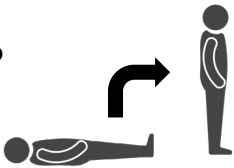
What determines stroke
volume?

MAP = mean arterial pressure
ICP = intracranial pressure
CO = cardiac output
SVR = systemic vascular resistance

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Physiology of Baroreceptor Response

What happens with positional change?



1

Blood pools in lower extremities → ↓ preload
→ ↓ CO

2

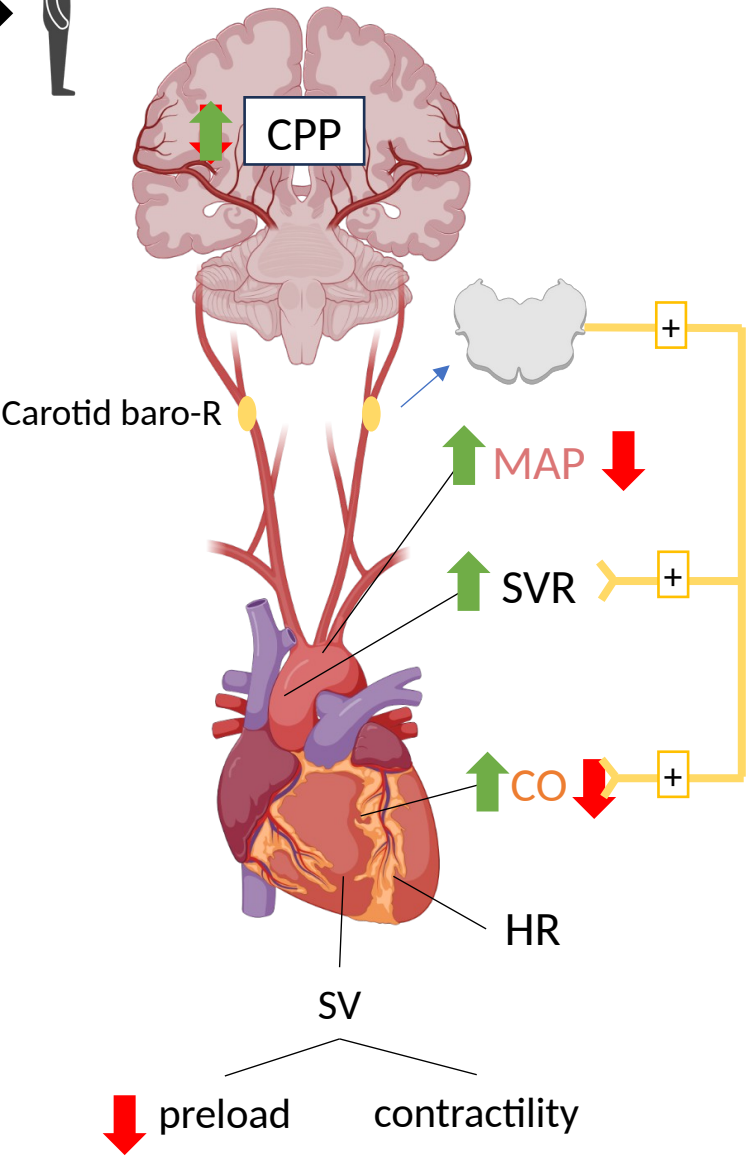
↓ CO → ↓ MAP → ↓ CPP

3

Baroreceptors sense ↓ MAP and alter
sympathetic and parasympathetic systems

4

↑ sympathetic tone → ↑ CO and SVR
→ ↑ MAP and CPP



MAP = mean arterial pressure
ICP = intracranial pressure
CO = cardiac output
SVR = systemic vascular resistance

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Classification - Orthostatic

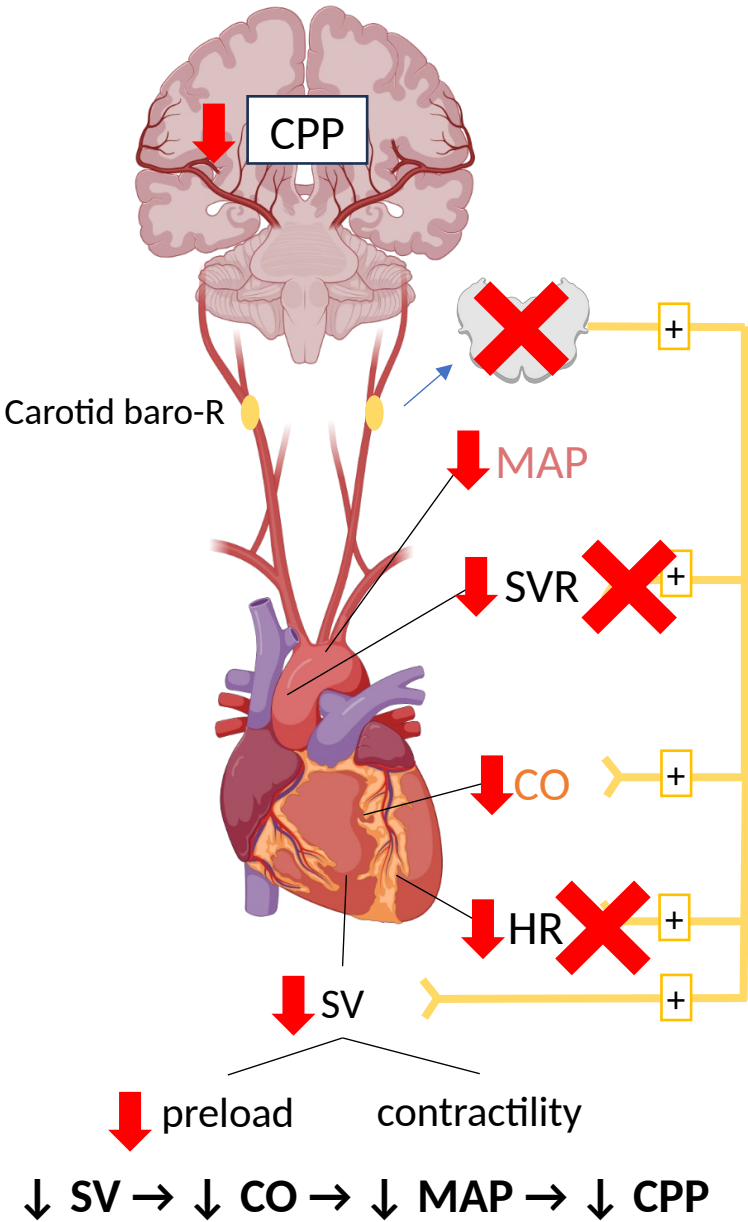
Postural hypotension due to an *insufficient* autonomic response

Definition

With postural change...
↓ SBP ≥ 20 or DBP ≥ 10
↓ SBP < 90 with symptoms

Orthostatic syncope is caused by *insufficient*...

Sympathetic tone	Autonomic dysfunction (neurodegenerative disease, DM)
SVR	Adrenal insufficiency Anti-hypertensives
HR	AVD Dihydropyridines (BB, CCB)
Preload (SV)	Diuretics Hypovolemia Venous pooling



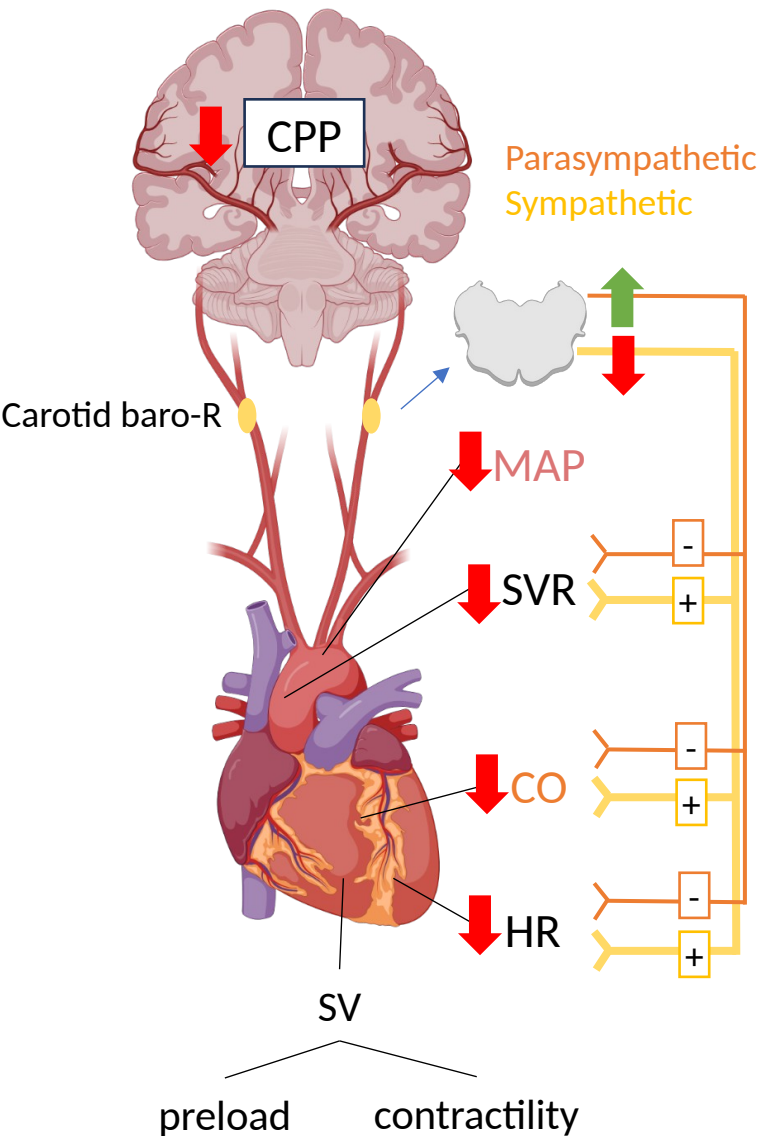
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Classification - Reflex Syncope

Definition	<i>Inappropriate</i> autonomic response to a physiologic stimulus or trigger
Differential	Vasovagal Situational Carotid sinus hypersensitivity

Mechanisms of reflex syncope

Cardio-inhibitory	↑ ↑ vagal tone → ↓ HR → ↓ CO
Vaso-depressive	↓ sympathetic tone → vasodilatation → ↓ SVR

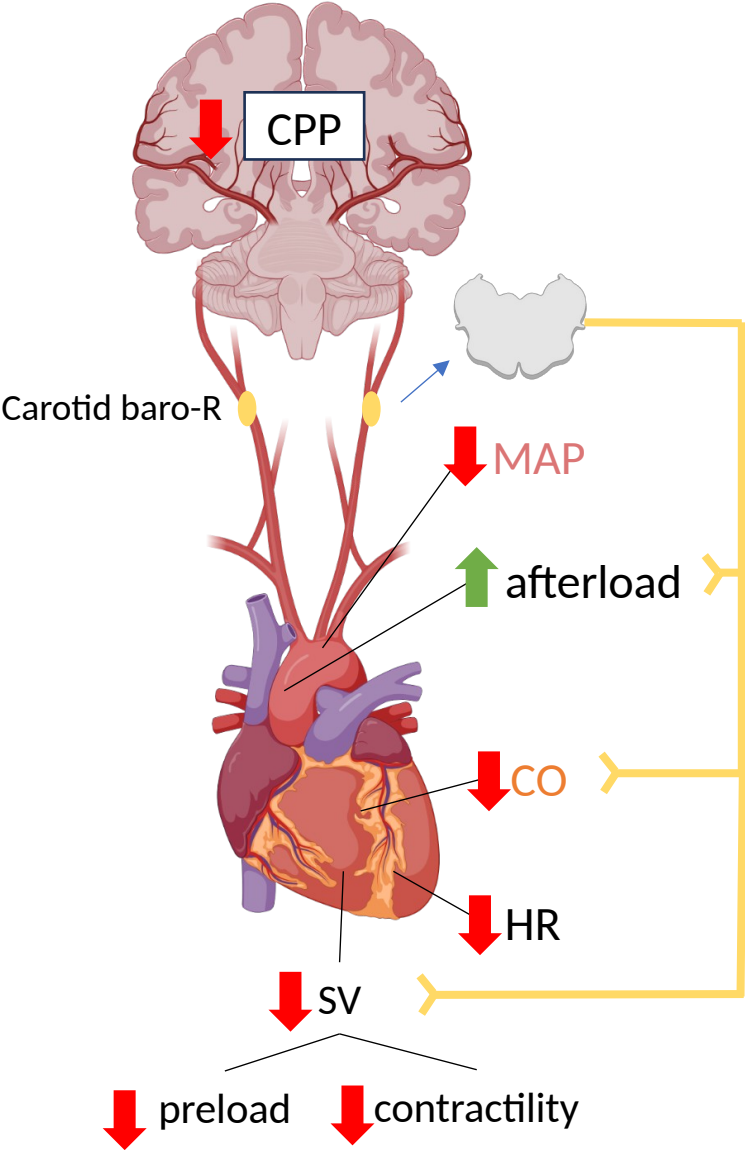


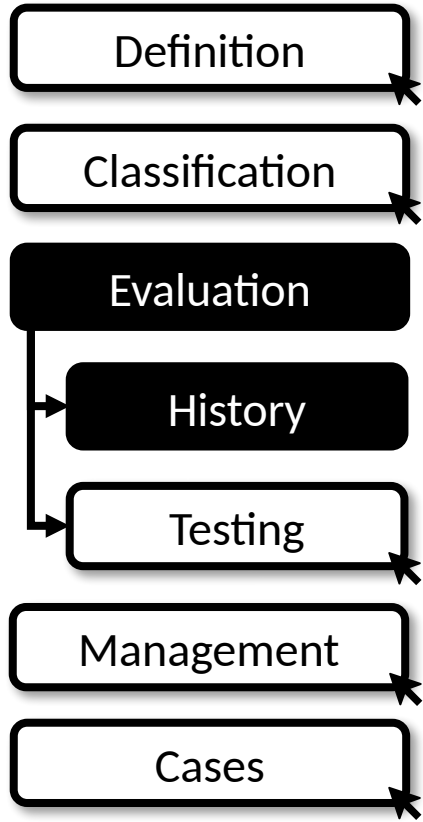
Classification - Cardiac

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Causes of cardiac syncope

↑ afterload	Aortic stenosis Differential HOCM
↓ contractility	MI Differential Ventricular arrhythmias
↓ HR	Heart block Differential Sinus node dysfunction
↓ preload	Tachyarrhythmias (SVT) Differential Tardus parvus, pulmonary HTN Aortic dissection





Evaluation - History

	Orthostatic	Reflex	Cardiac
Who	- Age: Old - PMH: Who? - Volumetric blood loss - DM, neuro-degenerative disease	- Age: All Who? - PMH: Generally healthy Who?	- Age: Old - PMH: Structural heart disease, CAD/MI, valvular disease Who? - FH: sudden cardiac death
Where How	Where? With postural change	Where? - Prolonged standing, urination/ defecation, coughing, vomiting - Stressful stimulus - Carotid stimulation	Where? - Supine, sitting, or at rest - With exertion
Prodrome	Prodrome? Lightheadedness, dizziness associated with postural change	Prodrome? Nausea, diaphoresis, pallor	Prodrome? - Chest pain, dyspnea, palpitations - No prodrome (i.e., unheralded syncope)

Patients with (type) require

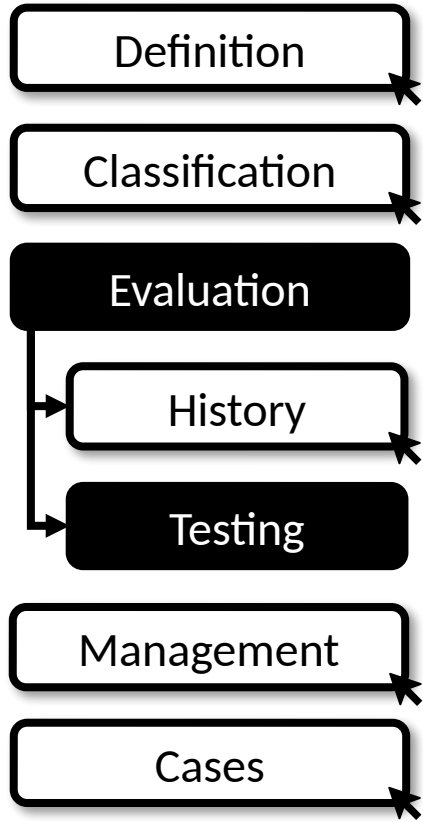
What are high risk features?

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Evaluation - Testing

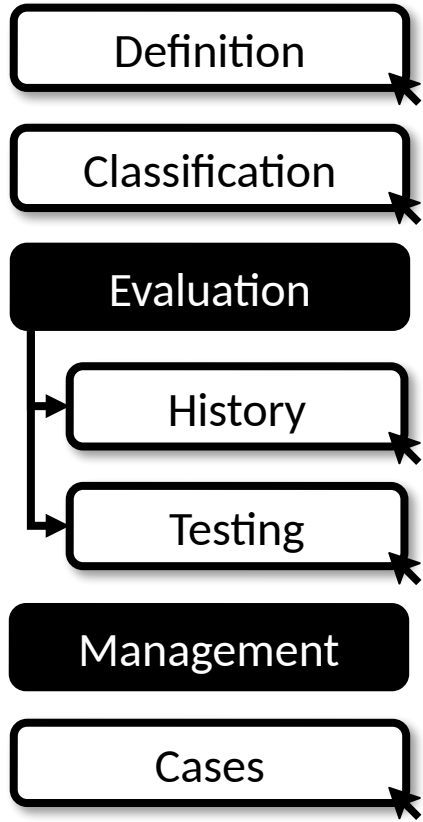
Evaluation

Orthostatic	Reflex	Cardiac
- History (signs of GIB) - Physical exam (SBP<90, HR<40, murmur) - Vital signs (orthostatics)	Standard evaluation ECG	- Acute ischemia - Structural heart disease - Bradyarrhythmias - Arrhythmogenic pattern (Brugada, long/short QT)
Head imaging (unless head injury or neuro symptoms)		
What testing is not routinely useful?		
Carotid imaging		
Patients with	What are high risk features?	e) require
What additional testing can help differentiate causes of syncope?		



Evaluation - Testing

	Orthostatic	Reflex	Cardiac
Evaluation	• History (signs of GIB) • Physical exam (SBP<90, HR<40, murmur) • Vital signs (orthostatics)		
Targeted tests	• BMP, CBC • O • ± autonomic function evaluation Special tests?	Special tests? diagnosis is uncertain)	• ECG • Acute ischemia • Structural heart disease • Bradyarrhythmias • Arrhythmogenic pattern (Brugada, long/short QT) • Telemetry / ambulatory ECG monitoring • TTE • Stress test (if exertional syncope) Special tests?



Evaluation - Testing

	Orthostatic	Reflex	Cardiac
Evaluation	- History (signs of GIB) - Physical exam (SBP<90, HR<40, murmur) - Vital signs (orthostatics) - ECG		
Targeted tests	- BMP, CBC - Orthostatic VS ± autonomic failure evaluation	Tilt table testing (if diagnosis is uncertain)	- Telemetry/ ambulatory ECG monitoring - TTE ± stress test (if exertional syncope)
Treatment	- Volume resuscitate - Stress management - Behaviors, counter pressure maneuvers	- Reassurance - Trigger avoidance - Counter pressure maneuvers	Consult Cardiology (treatment individualized based on etiology)
Second line treatment	- Mifegestonol - Fludrocortisone	Consider pacemaker (>40 yo, TTT) + frequent & unpredictable syncope)	

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Case 1

21-year-old presents for syncope.

VS: 36.5C, HR 82, BP 112/87

Exam: no murmurs, otherwise normal

What history is important to obtain?

- *Who*: history of depression on escitalopram
- *Where/how*: Occurred standing in line at the grocery store
- *Prodrome*: Clamminess, tunnel vision

What initial tests would you order?

EKG: normal sinus rhythm with normal intervals, no signs of chamber hypertrophy or prior ischemia.
Routine labs are not needed.

What is the most likely diagnosis?

Vasovagal syncope

Where would you treat this patient?

Outpatient

How would you treat this patient?

Provide reassurance

Case 2

56-year-old with HTN, chronic venous stasis presents with syncope. VS: 36.5C, HR 112, BP 112/87

Exam: no murmurs, dry mucus membranes, poor skin turgor.

What history is important to obtain?

- *Who*: history of HTN on lisinopril. Recent GI illness.
- *Where/how*: Occurred while standing up
- *Prodrome*: light-headedness and dizziness

What initial tests would you order?

Orthostatics: HR 112, BP 112/70 → HR 130, BP 85/40

ECG: sinus tachycardia

CBC normal, Na 132, K 3.2, bicarb 18, Cr 1.8, lactate 2.5

What is the diagnosis?

Orthostatic hypotension

Where would you treat this patient?

Urgent care or ED, consider observation depending on response to treatment

How would you treat this patient?

IVF resuscitation, hold BP medications

Definition

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Case 3

65-year-old with CAD s/p PCI (2 years prior), HFrEF 2/2 ischemic cardiomyopathy, HTN, DM type II presents with loss of consciousness + convulsions.

VS: 36.5C, HR 72, BP 112/87

Exam: normal heart rate but note occasional irregular beats, no murmurs, appears euvolemic

What history is important to obtain?

- *Who*: 65-year-old with significant cardiac history on furosemide, carvedilol, sacubitril-valsartan, spironolactone, empagliflozin, aspirin, metformin. History of episodic palpitations and lightheadedness x 2 weeks that come on at rest.
- *Where/how*: Occurred while watching TV when they became unresponsive x 20 s. Had witnessed whole-body convulsions. EMS arrived after they had regained consciousness. He denied any confusion.
- *Prodrome*: none

What initial tests would you order?

CBC, BMP. Troponin and BNP.

ECG

Orthostatic vital signs

Case 3 cont...

- Orthostatics: HR 72, BP 112/70 → HR 85, BP 115/65
- BMP: Na 132, K 3.2, Cl 110, Cr 0.8, BUN 15
- CBC: WBC 10.2, Hgb 12.8, plt 152k
- hsTroponin I: 140 → 100 ng/L
- ECG: normal sinus rhythm with Q waves in anterolateral leads, no BBB, frequent PVCs, no new ischemic change

What is the most likely cause of syncope?

Cardiac syncope. Further testing is needed to clarify diagnosis.

Where would you treat this patient?

In the hospital.

How would you further evaluate this patient?

TTE, telemetry

...TTE shows left ventricular wall motion abnormalities, LVEF 30%. Mild aortic stenosis. Telemetry with frequent symptomatic NSVT.

What is your next step in management?

Consult cardiology! Etiology is most likely NSVT.

Take-Home Points

1

Syncope = global cerebral hypoperfusion

Pathophysiology defines it and separates it from other transient loss of consciousness (seizure, hypoglycemia, TIA — the three big mimics).

2

Three classifications

Orthostatic hypotension (insufficient autonomic response), reflex (inappropriate response to a trigger), cardiac (low CO — the one that kills).

3

History does the heavy lifting

Who + where/how + prodrome + medications distinguish the three types in most patients.

4

Everyone gets four things

History, physical exam, orthostatic vitals, ECG. Everything beyond that is individualized; high-risk features → admit.

About This Optimized Edition

Original: “Syncope” — TeachIM.org. Hibshman M, Dietsche C; eds. Zhang Y, Fainstad B (Univ. of Colorado / Univ. of Washington). Published October 2023.

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Guideline currency (verified June 2026): 2017 ACC/AHA/HRS and 2018 ESC syncope guidelines remain the standing references; core content confirmed current.

Corrections in this edition: Case 3 orthostatic heart rate “HR 850” → “HR 85”; high-sensitivity troponin units “ng/dL” → “ng/L” (standard hs-cTn reporting).

Newer context (see lesson package): 2021 ESC pacing guideline + BIOSync strengthened support for dual-chamber (CLS) pacing in patients >40 with severe recurrent cardioinhibitory reflex syncope; cardioneuroablation is an emerging device-free option under study.

Companion files: Syncope_Lesson_Package.html (pre-work, teaching script, MCQs, case answers) · Syncope_Anki_Cards.txt